Tutorial Sheet 3

The Linear Quadratic Regulator

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Exercise 1. Consider the discrete-time, finite-horizon optimal control problem. That is, consider the system

$$\dot{x} = Ax + Bu\,,$$

and the cost functional

$$J(u, x_0) = \frac{1}{2}\sum_{k=0}^{N} x_k^\top Q x_k + u_k^\top R u_k \,\mathrm{d}t\,,$$

which we wish to minimise via the selection of the control input u.

(a) Show that the Bellman equation associated with the problem is

$$V(x, k) = \min_u \{\frac{1}{2}x^\top Q x + \frac{1}{2}u^\top R u + V(Ax + Bu, k+1)\}\,.$$

(b) Explaining your reasoning, state what boundary condition is the value function $V(x, k)$ subject to.

(c) Show that $V(x, k) = \frac{1}{2}xP(k)x$, where

$$P_k = Q + A^\top P_{k+1} A - A^\top P_{k+1} B (R + B^\top P_{k+1} B)^{-1} B^\top P_{k+1} A\,,$$

with

$$P_N = Q_N\,,$$

is a solution of the Bellman equation.

(d) The optimal control can be expressed as a feedback strategy. Namely, it can be expressed as

$$u = -K^\star x\,.$$

What is the expression for $K^\star$ (in terms of the system matrices and P_k)?

(e) Suppose the initial condition $x(0) = x_0$. What is the optimal cost-to-go from this initial condition?

(a) $V(x,k) = J^*(x)$

$$= \min_u \{ L(x, \mu(x)) + V(x_{k+1}, k+1) \}$$

$$= \min_u \{ L(x, \mu(x)) + V(f(x_k, u_k), k+1) \}$$

$$= \min_u \{ \tfrac{1}{2} x^T Q x + \tfrac{1}{2} u^T R u + V(Ax + Bu, k+1) \}$$

(b) Since the finite-horizon problem is solving by backward recursion, the boundary condition therefore is the terminal cost:

$$V(x, N) = \Phi(x, N) = \tfrac{1}{2} x_N^T Q_N x_N$$

(c) $V(x,k) = \min_u \{ \tfrac{1}{2} x^T Q x + \tfrac{1}{2} u^T R u + \tfrac{1}{2} (Ax + Bu)^T P_{k+1} (Ax + Bu) \}$

$$= \tfrac{1}{2} \min_u \{ x^T Q x + u^T R u + (x^T A^T + u^T B^T) P_{k+1} (Ax + Bu) \}$$

$$= \tfrac{1}{2} \min_u \{ x^T Q x + u^T R u + x^T A^T P_{k+1} A x + x^T A^T P_{k+1} B u$$
$$+ u^T B^T P_{k+1} A x + u^T B^T P_{k+1} B u \}$$

$$= \tfrac{1}{2} \min_u \{ x^T (Q + A^T P_{k+1} A) x + u^T (R + B^T P_{k+1} B) u$$
$$+ x^T A^T P_{k+1} B u + u^T B^T P_{k+1} A x \}$$

$$= \tfrac{1}{2} \min_u \{ x^T (Q + A^T P_{k+1} A) x + u^T (R + B^T P_{k+1} B) u$$
$$+ 2 x^T A^T P_{k+1} B u \}$$

$$= \tfrac{1}{2} \min_u D$$

$$\therefore \nabla_u D = 2 u^T (R + B^T P_{k+1} B) + 2 x^T A^T P_{k+1} B$$

for $\nabla_u D |_{u=u^*} = 0$, where $\nabla_u^2 D = 2(R + B^T P_{k+1} B) > 0$

$$u^* = -(R^T + B^T P_{k+1}^T B)^{-1} B^T P_{k+1}^T A x$$

$\because P = P^T > 0,\ R = R^T > 0$

$\therefore u^* = -(R + B^T P_{k+1} B)^{-1} B^T P_{k+1} A x = -Sx$

$\therefore V(x, k) = \frac{1}{2} x^T P_k x$

$= \frac{1}{2} x^T (Q + A^T P_{k+1} A) x + \frac{1}{2} x^T S^T (R + B^T P_{k+1} B) S x$

$- x^T A^T P_{k+1} B S x$

$\therefore P_k = Q + A^T P_{k+1} A + S^T (R + B^T P_{k+1} B) S - 2A^T P_{k+1} B S$

$= Q + A^T P_{k+1} A$

$+ A^T P_{k+1} B (R + B^T P_{k+1} B)^{-1} (R + B' P_{k+1} B)(R + B^T P_{k+1} B)^{-1} B^T P_{k+1} A$

$- 2 A^T P_{k+1} B (R + B^T P_{k+1} B)^{-1} B^T P_{k+1} A$

$= Q + A^T P_{k+1} A - A^T P_{k+1} B (R + B^T P_{k+1} B)^{-1} B P_{k+1} A$

where $V(x, N) = \frac{1}{2} x_N^T P_N x_N = \frac{1}{2} x_N^T Q_N x_N$

$\therefore P_N = Q_N$

(d) $K = (R + B^T P_{k+1} B)^{-1} B^T P_{k+1} A$

(e) $V(x_0, 0) = \frac{1}{2} x_0^T P_0 x_0$

where $P_0 = Q_0 + A^T P_1 A - A^T P_1 B (R_0 + B^T P_1 B)^{-1} B^T P_1 A$

Exercise 2. Consider a nonlinear system with dynamics

$$\dot{x} = f(x) + g(x)u\,,$$

where $x : \mathbb{R}^n$ and $u : \mathbb{R}^m$ denote the state and the control input, respectively, and $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^n$ are sufficiently smooth. Suppose we wish to design u to minimise the cost functional

$$J(u, x_0) = \frac{1}{2}\int_0^\infty q(x) + u^\top R u \,\mathrm{d}t\,,$$

where $q(x) > 0$ for all $x \neq 0$ and $q(0) = 0$, and $R = R^\top > 0$.

a) What is the Hamilton-Jacobi-Bellman equation associated with this infinite-horizon optimal control problem?

b) Show that $V(x) > 0$ for all $x \neq 0$ and $V(0) = 0$.

c) Show that $\dot{V}(x) < 0$ for all $x \neq 0$.

d) Consider now the observations made in (b) and (c) above. We will now consider what these imply *intuitively*.

   i) What do the observation in (c) above means in terms of the time-evolution of the function $V(x)$.

   ii) In light also of the observation made in point (b) above, what does this imply about the time evolution of the state x?

e) Suppose now that $f(x) = Ax$, $g(x) = B$ and $q(x) = x^\top Q x$, for some constant matrices A, B and $Q = Q^\top > 0$ of appropriate dimensions. This corresponds to a continuous-time, infinite-horizon optimal control problem. Suppose (A, B) is reachable.

   i) Show that $V = \frac{1}{2}x^\top P x$ is a solution of the Hamilton-Jacobi-Bellman equation obtained in (a), where $P = P^\top > 0$ satisfies the Algebraic Riccati Equation (ARE)

$$PA + A^\top P - PBR^{-1}B^\top P + Q = 0\,.$$

   ii) What is the optimal control associated with this LQR problem?

   iii) The ARE is quadratic in P and possesses multiple solutions. The matrices A, B and Q are such that it possesses a unique solution which is such that $P > 0$. Nevertheless, it also admits a solution $\bar{P} = \bar{P}^\top < 0$. Explain why this solution is not relevant in determining the optimal control.

(a) $H(x, u, \lambda) = \min_u \{ L(x, u) + \lambda^T \dot{x} \}$

$$= \min_u \{ \tfrac{1}{2} q(x) + \tfrac{1}{2} u^T R u + \lambda^T (f(x) + g(x) u) \}$$

$$\therefore \left. \frac{\partial H}{\partial u} \right|_{u = u^*} = u^{*T} R + \lambda^T g(x) = 0 \quad \text{where} \quad \frac{\partial^2 H}{\partial u^2} = R > 0$$

$$\therefore u^* = -(R^T)^{-1} g^T \lambda = -R^{-1} g^T \lambda$$

$$\therefore H^*(x, \lambda) = \tfrac{1}{2} q + \tfrac{1}{2} \lambda^T g R^{-1} R R^{-1} g^T \lambda + \lambda^T (f - g R^{-1} g^T \lambda)$$

$$= \tfrac{1}{2} q + \tfrac{1}{2} \lambda^T g R^{-1} g^T \lambda + \lambda^T f - \lambda^T g R^{-1} g^T \lambda$$

$$= \tfrac{1}{2} q - \tfrac{1}{2} \lambda^T g R^{-1} g^T \lambda + \lambda^T f$$

$\therefore$ HJB equation for finite-horizon optimum control problem is

$$0 = H^*\left(x, \frac{\partial V}{\partial x}\right) = \tfrac{1}{2} q - \tfrac{1}{2} \left(\frac{\partial V}{\partial x}\right)^T g R^{-1} g^T \left(\frac{\partial V}{\partial x}\right) + \left(\frac{\partial V}{\partial x}\right)^T f$$

(b) Since $L(x, u) = \tfrac{1}{2} q(x) + \tfrac{1}{2} u^T R u > 0$, $V(x, t) > 0$ for all $x \neq 0$

$V(0) = 0$ as there is no cost-to-go at the $\boxed{t = 0}$ $x = 0$

(c) $$\dot{V} = \frac{\partial V}{\partial t} = \left(\frac{\partial V}{\partial x}\right)^T \dot{x} = \left(\frac{\partial V}{\partial x}\right)^T (f + g u^*) = \left(\frac{\partial V}{\partial x}\right)^T \left(f - g R^{-1} g^T \left(\frac{\partial V}{\partial x}\right)\right)$$

$$= -\tfrac{1}{2} q + \tfrac{1}{2} \left(\frac{\partial V}{\partial x}\right)^T g R^{-1} g^T \left(\frac{\partial V}{\partial x}\right) - \left(\frac{\partial V}{\partial x}\right)^T g R^{-1} g^T \left(\frac{\partial V}{\partial x}\right)$$

$$= -\left\{ \tfrac{1}{2} q + \tfrac{1}{2} \left(\frac{\partial V}{\partial x}\right)^T g R^{-1} g^T \left(\frac{\partial V}{\partial x}\right) \right\} < 0$$

(will strictly decreased with time)

(d) (i) the function is positive definite and has a minimum at the origin

(ii) as time increases, the value function must converge to the minimum

(e) (i) $$\frac{\partial V}{\partial x} = x^T P = P x$$

$$\therefore u^* = -R^{-1} g^T P x = -R^{-1} B^T P x$$

$$\therefore\ 0 = \frac{1}{2}q - \frac{1}{2}\left(\frac{\partial V}{\partial x}\right)^T g R^{-1} g^T \left(\frac{\partial V}{\partial x}\right) + \left(\frac{\partial V}{\partial x}\right)^T f$$

$$= \frac{1}{2}x^T Q x - \frac{1}{2}x^T P B R^{-1} B^T P x + x^T P A x$$

$$\therefore\ 2PA - PBR^{-1}B^T P + Q = 0$$

$$\therefore\ PA + A^T P^T - PBR^{-1}B^T P + Q = 0$$

(ii) $u^* = -R^{-1}B^T P x$

(iii) $A_{cl} = A + BK = A - BR^{-1}B^T P$

If $P < 0$, $\lambda(A_{cl})$ is impossible to lies at the open LHP i.e. solution is unstable

Exercise 3. Consider the simplified model of a satellite in orbit, that we considered also in Tutorial Sheet 1. Namely,

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{1}{M_I} \end{bmatrix} u(t),$$

where M_I is the satellite's moment of inertia, $x = [\theta\,\omega]^\top$ is the state of the system, with $\theta(t) = \theta(t)$ representing the angular deviation from the desired orientation and $\omega(t) = \dot{\theta}(t)$ representing the angular velocity, and the control input $u(t)$ is the torque applied by satellite's thrusters. Suppose $M_I = 1$ kg m^2.

Suppose now that we wish to design the control input u to minimise a cost functional of the form

$$J = \int_0^\infty \big(x(t)^\top Q x(t) + u(t)^\top R u(t)\big)\, dt\,.$$

This is essentially an "infinite-horizon version" of the cost you considered in Tutorial Sheet 1. There you were asked discuss how to assign the different elements of the cost functional.

a) What is the Algebraic Riccati Equation (ARE) associated with this problem?

b) What is the optimal control, in terms of the solution of the ARE?

c) What is the optimal cost-to-go, in terms of the solution opf the ARE and the initial condition $x(0) = x_0$?

d) In the remainder of this question, you are asked to pick different values for the various components of Q and R. Note that these have to be such that the problem admits a (unique) solution, as discussed in the lecture handouts. Suppose also that the initial condition $x_0 = 5$.

   For each of your selections, to do the following:

   i) Solve the algebraic Riccati equation (ARE). In Matlab this can be easily done using the function icare.

   ii) Integrate. e.g. using the function ode45 in Matlab, the system dynamics in closed-loop with the optimal control, over a time horizon $[0, T]$, where you should pick T such that the state has time to converge. Note that the value of T may vary, depending on your choice of Q and R.

   iii) Using your solution to the preceding part, plot the time histories of the state and of the control input.

iv) Compute the cost incurred up to time $t = T$ numerically and verify that this coincides with the optimal cost-to-go.

Hint: An easy way to compute the cost is to add an additional element to the state vector that corresponds to the cost incurred up to a given time. Then, the numerical integration can be performed simultaneously as you integrate the system dynamics, by noting that $\dot{J} = \frac{1}{2}x^\top Qx + u^\top Ru$ and that the cost incurred at time t=0 is zero (giving the initial condition required to solve the above differential equation).

(a) $\dot{x} = Ax + Bu$

$$H(x, u, \lambda) = \min_u \{x^T Q x + u^T R u + \lambda^T (Ax + Bu)\}$$

$$\therefore \left.\frac{\partial H}{\partial u}\right|_{u=u^*} = 2u^T R + \lambda^T B = 0 \quad \text{where} \quad \frac{\partial^2 H}{\partial u^2} = 2R > 0$$

$$\therefore u^* = -\frac{1}{2} R^{-1} B^T \lambda$$

$$\therefore H^*(x, \lambda) = x^T Q x + \frac{1}{4} \lambda^T B R^{-1} R R^{-1} B^T \lambda + \lambda^T (Ax - \frac{1}{2} B R^{-1} B^T \lambda)$$

Let $\lambda = \frac{\partial V}{\partial x}$, $V = x^T P x$, $P = P^T > 0$ $\quad \therefore \frac{\partial V}{\partial x} = 2Px$

$$\therefore 0 = H^*(x, \frac{\partial V}{\partial x})$$

$$= x^T Q x + x^T P B R^{-1} B^T P x + 2x^T P (Ax - B R^{-1} B^T P x)$$

$$= x^T (Q + P B R^{-1} B^T P + 2PA - 2 P B R^{-1} B^T P) x$$

$$= x^T (Q + 2PA - P B R^{-1} B^T P) x$$

$$\therefore Q + PA + A^T P - P B R^{-1} B^T P = 0$$

(b) $u^* = -R^{-1} B^T P x$

(c) $V(x_0, 0) = x_0^T P x_0$

(d) $Q = Q^T$ and $R = R^T > 0$

Therefore, let

① $Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $R = 1$

② $Q = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$, $R = 0.5$

③ $Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $R = 20$

see MATLAB